Machine Learning 2 (APM_50441_EP/DSAIB - 2026-2027). PC6

Exercise 1 (Policy evaluation and Bellman operator)

Consider the following Markov Decision Process (MDP) with for $t \in \mathbb{N}$:

- a state S_t valued in a space of states $(s_i)_{i\in I}$ and Card$(I) < \infty$;
- an action A_t valued in a space of actions $(a_j)_{j\in J}$ and Card$(J) < \infty$;
- a reward R_t valued in a space of rewards $(r_k)_{k\in K}$ and Card$(K) < \infty$.

We assume that for any $(i, j, k, i') \in I \times J \times K \times I$, the dynamics of the system is defined by the known transition law

$$p(s_{i'}, r_k|s_i, a_j) = \mathbb{P}\big(S_{t+1} = s_{i'},\, R_{t+1} = r_k \,\big|\, S_t = s_i,\, A_t = a_j\big).$$

Now we would like to choose a policy π, which chooses the action A_t at each t. For any π, we will denote $\mathbb{P}_\pi$ the law of the resulting process and, when π is a stationary Markov policy (the law of A_t only depends on S_t), P_π the transition matrix of the Markov chain (S_t). To choose an "optimal" policy π and with a fixed $\gamma \in (0, 1)$, for any π, we define the following Value function by

$$v_\pi(s) = \mathbb{E}_\pi\left[\sum_{t=0}^{+\infty} \gamma^t\, R_{t+1} \,\middle|\, S_0 = s\right] = \sum_{t=0}^{+\infty} \gamma^t\, \mathbb{E}_\pi[R_{t+1} \,|\, S_0 = s]\,,$$

for any $s \in (s_i)_{i\in I}$. For a stationary Markov policy π, we also define the expected reward $r_\pi(s) = \mathbb{E}_\pi[R_{t+1} \,|\, S_t = s]$ and the Bellman Operator, acting on any function v of the states, by

$$\mathcal{T}_\pi v(s) = r_\pi(s) + \gamma \sum_{s'\in(s_i)_{i\in I}} \mathbb{P}_\pi(S_{t+1} = s' \,|\, S_t = s)\; v(s'), \quad \text{i.e.} \quad \mathcal{T}_\pi v = r_\pi + \gamma\, P_\pi\, v.$$

1. Prove that if π is a stationary Markov policy and $0 < \gamma < 1$, then v_π is the only solution of $v = \mathcal{T}_\pi v$, i.e.

   $$v = r_\pi + \gamma P_\pi v,$$

   and $v_\pi = (\mathrm{Id} - \gamma P_\pi)^{-1} r_\pi$.

2. For any v_0, define $v_{n+1} = \mathcal{T}_\pi v_n$. Prove that

   $$\lim_{n\to\infty} v_n = v_\pi$$

   and

   $$\|v_n - v_\pi\|_\infty \le \gamma^n \|v_0 - v_\pi\|_\infty$$

3. Prove that

   $$\|v_n - v_\pi\|_\infty \le \frac{\gamma}{1-\gamma} \|v_n - v_{n-1}\|_\infty$$

4. Finally, prove that if $v_0 \ge \mathcal{T}_\pi v_0$ (respectively $v_0 \le \mathcal{T}_\pi v_0$) then $v_0 \ge v_\pi$ (respectively $v_0 \le v_\pi$) and v_n converges monotonically to v_π.

Exercise 2 (Stochastic Algorithm)

Assume that $(\mathcal{F}_i)_{i\in\mathbb{N}}$ is a filtration, i.e. a non-decreasing sequence of σ-algebras, and $(X_i)_{i\in\mathbb{N}}$ is a sequence of random variables adapted to $(\mathcal{F}_i)$. Assume that $\mathbb{E}[X_i \mid \mathcal{F}_{i-1}] = \mu$ and $\mathbb{V}\mathrm{ar}\,[X_i \mid \mathcal{F}_{i-1}] \le \sigma^2$ for $i \in \mathbb{N}^*$. Let $(\alpha_i)_i$ be a sequence of real numbers in $[0, 1]$ and define $(M_n)_n$ such that

$$M_n = M_{n-1} + \alpha_n\,(X_n - M_{n-1}) \quad \text{for } n \in \mathbb{N}^* \quad \text{and } M_0 = 0.$$

1. Prove that if $\sum_{i=1}^{\infty} \alpha_i = \infty$ and $\sum_{i=1}^{\infty} \alpha_i^2 < \infty$ then $M_n \xrightarrow[n\to\infty]{} \mu$ in L^2, i.e. $\mathbb{E}\big[(M_n - \mu)^2\big] \xrightarrow[n\to\infty]{} 0$.

2. Prove that if $\alpha_i = \alpha \in (0,1)$ for any $i \in \mathbb{N}$, then $\operatorname{limsup} \mathbb{E}\left[\left(M_n - \mu\right)^2\right] \leq \alpha\sigma^2$.

Exercise 3 (Optimal Bellman operator and policy iteration)

In the setting of the first exercise, define the action-value function of a stationary policy $q_\pi(s,a) = \mathbb{E}_\pi\left[\sum_{t\geq 0}\gamma^t R_{t+1}\middle| S_0 = s, A_0 =$ the optimal Bellman operator $\mathcal{T}^\star v(s) = \max_a\left(r(s,a) + \gamma\sum_{s'} p(s'|s,a)v(s')\right)$, where $r(s,a)$ is the expected reward and $p(s'|s,a)$ the transition probability, and the *greedy* policy with respect to a function v, which chooses in each state an action achieving this maximum.

1. Prove that $\mathcal{T}^\star$ is a γ-contraction for $\|\cdot\|_\infty$. Deduce that it has a unique fixed point $v^\star$, and that value iteration $v_{n+1} = \mathcal{T}^\star v_n$ converges to it at the rate γ^n.

2. *Policy improvement.* Let π be a stationary policy and π' the greedy policy with respect to v_π. Prove that $v_{\pi'} \geq v_\pi$. Deduce that policy iteration (evaluate, then improve) stops after a finite number of steps on an optimal policy.

3. *From values to policies.* Let q be an approximation of the optimal action-value $q^\star$ with $\|q - q^\star\|_\infty \leq \varepsilon$, and π the greedy policy with respect to q. Prove that $\|v_\pi - v^\star\|_\infty \leq \frac{2\varepsilon}{1-\gamma}$.

Exercise 4 (Policy gradient)

We consider episodes $\tau = (S_0, A_0, R_1, \ldots, S_T)$ generated by a parametric policy $\pi_\theta(a|s)$, and $J(\theta) = \mathbb{E}_{\tau\sim\pi_\theta}[G(\tau)]$ with $G(\tau) = \sum_{t=0}^{T-1} R_{t+1}$ (no discount).

1. *REINFORCE.* Using $\nabla_\theta p_\theta = p_\theta \nabla_\theta \log p_\theta$, prove that $\nabla_\theta J(\theta) = \mathbb{E}[G(\tau)\sum_t \nabla_\theta \log \pi_\theta(A_t|S_t)]$.

2. *Reward-to-go and baseline.* Prove that $\mathbb{E}[R_{t'+1}\nabla_\theta \log \pi_\theta(A_t|S_t)] = 0$ for $t' < t$, and that $\mathbb{E}[b(S_t)\nabla_\theta \log \pi_\theta(A_t|S_t)] = 0$ for any function b of the state. Deduce that $G(\tau)$ can be replaced by the reward-to-go $G_t = \sum_{t'\geq t} R_{t'+1}$ minus a baseline $b(S_t)$ without biasing the gradient. Why is it useful?

3. *Softmax policy.* For a tabular softmax policy $\pi_\theta(a|s) = e^{\theta_{s,a}} / \sum_b e^{\theta_{s,b}}$, compute $\nabla_{\theta_{s,\cdot}} \log \pi_\theta(a|s)$.

Exercise 5 (Reinforcement learning for language models) 1. *KL-regularized optimum (RLHF and DPO).* For a prompt x, a reference policy $\pi_{\text{ref}}(y|x)$, a reward $r(x,y)$ and $\beta > 0$, show that the policy maximizing $\mathbb{E}_{y\sim\pi(\cdot|x)}[r(x,y)] - \beta\ \mathrm{KL}(\pi(\cdot|x), \pi_{\text{ref}}(\cdot|x))$ is $\pi^\star(y|x) = \pi_{\text{ref}}(y|x)e^{r(x,y)/\beta}/Z(x)$. Deduce that $r(x,y) = \beta\log\frac{\pi^\star(y|x)}{\pi_{\text{ref}}(y|x)} + \beta\log Z(x)$, and that with the Bradley–Terry model $\mathbb{P}(y_w \succ y_l) = \sigma(r(x,y_w) - r(x,y_l))$ the likelihood of human preferences can be written directly in terms of the policy (DPO).

2. *Group baselines (GRPO).* For a prompt, G answers are sampled with rewards $r_1, \ldots, r_G$ (e.g. 1 if the answer is verified correct). The leave-one-out baseline uses $b_i = \frac{1}{G-1}\sum_{j\neq i} r_j$; GRPO uses the group mean $\bar{r}$ (and divides by the standard deviation). Show that the leave-one-out baseline keeps the policy gradient unbiased, and that $r_i - \bar{r} = \frac{G-1}{G}(r_i - b_i)$.

3. *PPO clipping.* PPO maximizes $\mathbb{E}[\min(\rho A, \mathrm{clip}(\rho, 1-\epsilon, 1+\epsilon)A)]$, where $\rho = \pi_\theta(a|s)/\pi_{\theta_k}(a|s)$ and A is the advantage. For $A > 0$ and $A < 0$, describe the values of ρ for which the gradient of the objective is zero. Why is it a "pessimistic" bound?

4. *Back-of-the-envelope.* (a) An agent performs a task in 20 steps, each step being correct with probability 0.95 independently: what is the probability that the whole task succeeds? With 0.99? (b) A token costs 4 times less than last year, but a reasoning model uses 250 times more tokens per answer: how does the cost of an answer change?

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